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Text Mining and Design Optimization Study Based on Online Reviews of Cultural and Creative Products

Xin TIAN [a,b,1] Shijian LUO[a] Hui CHENG[c] Changao LIU[a]

[a] *Department of Industrial Design, College of Computer Science and Technology, Zhejiang University, Hangzhou 310027, China*
[b] *College of Management, Guizhou University, Guiyang 550025, China*
[c] *Zhejiang University of Finance and Economics Dongfang College*

Abstract. This paper systematically analyzes consumer reviews of cultural and creative products from the Palace Museum using text mining methods. The study aims to uncover consumer focal points regarding product design and to provide data support and decision-making references for design optimization. Through in-depth analysis from four perspectives—high-frequency word analysis, sentiment classification, feature keyword extraction, and topic extraction—the study reveals that consumer reviews primarily focus on aspects such as appearance design, gift-giving suitability, and cultural symbolism. Additionally, logistics efficiency, price rationality, and customer service quality are also significant factors influencing consumer satisfaction.

Keywords. Cultural and creative products; consumer reviews; text mining; high-frequency word analysis; sentiment classification; design optimization

1. Introduction

As a fusion of cultural creativity and tangible goods, cultural and creative products combine both creative and commercial attributes, offering consumers a unique consumption experience [1]. With the rise of cultural confidence and increased consumption levels in China, the market for cultural and creative products has experienced rapid growth [2]. Online reviews reflect consumer preferences for various product features [3], uncovering real consumer needs and satisfaction levels, thereby providing valuable data support for the design improvement of cultural and creative products. Existing research primarily focuses on aspects such as the market value [4], design concepts [5], and consumer behavior [6] related to cultural and creative products. However, few studies have leveraged e-commerce big data to explore consumer focal points through text mining of consumer reviews. Therefore, this study conducts a multi-dimensional analysis of consumer reviews of cultural and creative products from the Palace Museum, aiming to reveal consumer focal points and sentiment tendencies, thus offering insights for the design optimization of cultural and creative products.

[1] Corresponding Author: Xin TIAN. Email: 9204290@qq.com

2. Methods and data

2.1. Methods and Sample Selection

Text mining refers to the data mining of unstructured text, including methods such as sentiment analysis, word frequency analysis, and semantic network analysis. Scholars both domestically and internationally have widely applied it to review analysis [7-9]. For instance, Zhu Dongxiao et al. [10] analyzed online reviews from museum visitors, using the LDA model for topic identification and employing semantic network analysis to explore dimensions of visitor experience in reviews, ultimately establishing a five-dimensional visitor experience model: "basic facilities," "cultural and creative surroundings," "exhibit display," "interactive media," and "interpretation services." Similarly, Gong Yulu et al. [11], based on perceived value theory, conducted text mining on online reviews of cultural and creative products from the National Library of China, exploring the relationship between perceived value and consumer needs across four dimensions: emotion, society, quality, and price. Compared with market research and expert interviews, text mining provides a more efficient and convenient way to identify consumer focal points and preferences, saving time, effort, and resources [12].

In this study, Python and the Jieba library were used to create a word cloud evolution map of textual data, analyzing the reasons for consumer satisfaction or dissatisfaction with cultural and creative products. Taobao rankings show that the Palace Museum's Taobao store has nearly 9.5 million followers, with 742 products on sale across more than a dozen categories and annual sales exceeding 1 billion yuan, leading the museum-based cultural and creative product market in terms of followers, products, and sales [13]. The Palace Museum's cultural and creative products have a wide audience, and its official online store has a relatively standardized user review system. Therefore, consumer reviews from the Palace Museum's flagship store serve as the sample data for this study.

2.2. Data collection and processing

The data collection and processing procedures for this study include the following steps.

- First, online reviews related to the Palace Museum’s cultural and creative products were obtained from the official website (URL). The review data spans from November 2021 to November 2023, sorted by sales volume from high to low, covering basic information on 190 products. A total of 11,035 raw reviews were collected, with each review including six key elements: username, product name, sales volume, review date, review content, and response content. An initial analysis revealed that positive reviews significantly outnumber negative ones, with 8,931 positive reviews labeled as Dataset A, and 2,104 negative reviews labeled as Dataset B. Table 1 presents some sample raw data from Datasets A and B.

Table 1. Part of raw review data

Data sets	Nicknames	Comments
Review set A	Preferred **Freeze	I like this kind of literary style of cultural and creative products, gift is not cheesy.
	Star**9	The stuff is even more perfect than I imagined, very powerful, worth getting.
Review set B	t**0	A bit poorly packaged
	t**9	Not really worth the money

- In the second step, entries containing "This user did not leave a review" and empty reviews ("") were filtered out, resulting in a final dataset of 10,202 reviews. Additionally, leading and trailing spaces, as well as special characters, were removed from the reviews, and newline characters were replaced with spaces to facilitate further processing.
- The third step focused on preprocessing the reviews, which included segmenting the text into clauses, removing stop words, and performing word segmentation. The stop word lists were derived from the Harbin Institute of Technology and Sichuan University, while the word segmentation was performed using the Jieba library, which is specifically designed for Chinese text processing. The research process is illustrated in Figure 1.

3. Text comment analysis

3.1. Commentary high-frequency word analysis

After data preprocessing, the ROST Content Mining 6.0 software, developed by Wuhan University, was used to perform high-frequency word analysis on the review data, generating a high-frequency word statistics table (see Table 2). Table 2 reflects the key areas of consumer attention toward cultural and creative products, providing data support for further analysis of consumer needs and design optimization.

Table 2. Consumer Review High Frequency Vocabulary

High Frequency Words	Word Frequency	High Frequency Words	Word Frequency	High Frequency Words	Word Frequency
Good-looking	1874	Satisfied	464	Express	202
Friends	916	Workmanship	431	Quality	186
Beautiful	911	Give it to	377	Design	175
Exquisite	793	Good review	312	Texture	174
Packaging	759	Shipping	292	Fit	161
Give away	718	Logistics	273	Services	160
Quality	670	Exquisite	267	Speed	155
Gift	650	Very fast	246	Moral	153
The Forbidden	519	Material	240	Imagine	150
City		Appearance	225	Color	145
Cute	479				

Consumer evaluations of cultural and creative products primarily focus on aspects such as appearance, quality, social value, cultural symbolism, and brand. Words like "beautiful" and "pretty" rank highly, indicating that the visual appeal of the product is a primary concern for consumers. Following these are terms like "friend," "gift," and "present," which suggest that gifting is a common purpose for purchasing these products, highlighting the importance of considering social attributes in product design. Words such as "meaning" and "imagination" indicate that consumers value the cultural connotations and symbolism of the product. In design, there is potential to further explore and integrate traditional cultural elements to enhance the product's cultural depth and appeal.

3.2. Emotional analysis

This paper reveals the emotional tendencies of consumer behavior through sentiment analysis of reviews on cultural and creative products.

Table 3. Statistical Results of Consumer Sentiment Classification

Categories	Number	Ratio
Positive	8098	79.4%
Negative	2104	20.6%

The results of the sentiment analysis are presented in Table 3, indicating that positive emotions dominate user comments, accounting for 79.4% of the total. In contrast, negative emotional comments make up 20.6%. This suggests that the majority of consumers hold a positive view of the design of cultural and creative products and express satisfaction with their purchases.

3.3. Feature Keyword Analysis

Based on the segmentation results, word cloud maps (Figures 1 and 2) are generated to visualize the characteristic keywords of user comments in different datasets. The screening of key information is realized with the help of TF-IDF [14] (Term Frequency-Inverse Document Frequency) values, and the weight of each keyword is measured based on its TF-IDF score in the comments. In the key information extraction session, the first task is to count the frequency of each term in the comments, i.e., Term Frequency (TF), to present the density of terms in the comments. Following this, the inverse document frequency (IDF) is accounted for to judge how common the terms are in the overall corpus. By multiplying TF and IDF, the resulting TF-IDF value reflects the importance of the term in the comments, which is denoted by w_{ij}[15]. The specific calculation is shown in Equation (1)

$$w_{ij} = TF_{ij} \times IDF_{ij} = TF_{ij} \times log\left(\frac{N}{DF_j}\right) \quad (1)$$

In this context, TF_{ij} and IDF_{ij} denote the frequency of occurrence and the inverse document frequency for the characteristic term j within the i-th post, respectively. Herein, N signifies the aggregate count of all comments, while DF_j corresponds to the quantity of comments encompassing the particular feature word j [15] .



Figure 1. Commentary Set A Word Cloud Summary Chart

From Figure 1, it is evident that Dataset A contains a substantial number of positive evaluation words related to appearance design, aesthetics, and quality, including terms like "beautiful," "good," "friend," "exquisite," "quality," and "gift." This reflects users' recognition of the design of cultural and creative products and their success in visual appeal. Additionally, the frequent occurrences of "friend" and "gift" reveal that users tend to purchase these products as gifts for friends and family.

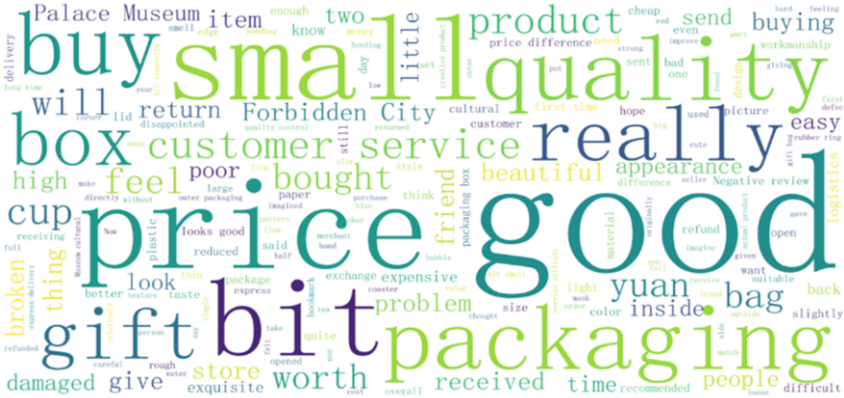


Figure 2. Commentary Set B Word Cloud Summary Chart

Figure 2 illustrates that dissatisfaction in Dataset B primarily focuses on aspects such as "price," "small," "packaging," "quality," and "customer service." The mention of "packaging" indicates that there is room for improvement in user satisfaction regarding packaging design. In design optimization, greater attention can be paid to the practicality and aesthetic appeal of packaging to enhance the consumer purchasing experience and overall satisfaction.

3.4. LDA-based comment thread extraction

Word clouds can reveal the main product features that buyers of cultural and creative products focus on, but they cannot display the relationships between these factors. To explore the thematic distribution within the data, this study employs the LDA[16](Latent Dirichlet Allocation) topic modeling algorithm to mine the implicit themes in the review data. As a generative probabilistic model for discrete datasets, such as text corpora[17], the LDA topic algorithm has distilled consumer reviews of the Palace Museum's cultural and creative products into eight distinct themes and their corresponding keywords (see Table 4).

LDA topic algorithm uses Gibbs Sampling for sampling:

$$p(z_i = k \mid \vec{\mathbf{z}}_{\neg i}, \vec{\mathbf{w}}) \propto \frac{n_{m,\neg i}^{(k)} + \alpha_k}{\sum_{k=1}^{K} (n_{m,\neg i}^{(k)} + \alpha_k)} \cdot \frac{n_{k,\neg i}^{(t)} + \beta_t}{\sum_{t=1}^{V} (n_{k,\neg i}^{(t)} + \beta_t)} \quad (2)$$

As shown in Equation (2), where K is the number of topics, V is the number of words, α is the K-dimensional vector of positive reals and β is the V-dimensional vector of positive reals. The p represents the conditional distribution corresponding to any coordinate axis i in the n-th word of the m-th article, excluding the word with subscript i.

Table 4. Classification Examples of subject categories and keywords

Subject type	Partial keywords
Design Attributes	Good-looking, exquisite, beautiful, exquisite, appearance, beautiful, shape, perfect, high value, stunning
Emotional Attributes	Like, surprise, intention, praise, satisfaction, good, cure, decompression, happy, not to give up
Social Attributes	Friend, girlfriend, give, gift, gift, gift, send, friend, gift, gift, gift box, festival
Quality Attributes	Quality, material, quality, workmanship, texture, fine, feel, exquisite, featured, carefully
Aesthetic Attributes	Literary, festive, high, elegant, showy, courtly, national style, ancient style, classical, peaceful atmosphere
Cultural Attributes	Cultural heritage, sentiment, traditional culture, cultural value, Chinese characteristics, cultural relics, cultural significance, cultural expression, moral, imagination
Functional Attributes	Functionality, usefulness, practicality, stamp, collection, writing, protection, decoration, wishes, ornamentation
Service Attributes	Shipping, soon, logistics, received, packaging, speed, attitude, service, courier, shipment

According to Table 4, user comments on cultural and creative products are primarily categorized into eight themes. Consumers place the highest importance on design attributes, followed by emotional satisfaction and social attributes. Quality, craftsmanship, aesthetics, culture, and functionality are also noted, while service attributes are of slightly lesser importance.

Design attributes emerge as the most critical theme, indicating that the popularity of cultural and creative products largely relies on clever design. Additionally, the emotional satisfaction provided by these products is highly valued, with frequent terms such as "like," "surprise," "satisfaction," "healing," and "stress relief" reflecting consumers' emotional needs. High-frequency words related to the social attributes of these products, including "friend," "gift," and "present," demonstrate their significance in social gifting scenarios, serving as important mediums for establishing connections with friends, family, and elders.

3.5. Analysis of emotional factors

- Positive topic

	Exquisite	Moral	Quality	Design	Friend	Favorite	Present	Good-looking	Gift	Workmanship	Logistics	Functionality	Texture	Culture	Jubilation
Exquisite	735	22	40	15	65	151	15	117	36	103	9	7	9	7	11
Moral	22	141	9	5	20	33	2	27	17	21	3	3	4	1	7
Quality	40	9	540	17	28	113	13	89	8	42	56	11	11	3	3
Design	15	5	17	128	8	43	3	18	4	12	2	6	4	5	2
friends	65	20	28	8	767	388	6	127	117	27	21	3	11	4	8
Favorite	151	33	113	43	388	1789	10	249	122	90	47	11	33	11	5
Present	15	2	13	3	6	10	103	13	5	2	3	1	1	1	1
Good looking	117	27	89	18	127	249	13	1670	58	52	30	11	38	8	8
Gifts	36	17	8	4	117	122	5	58	386	10	11	1	8	3	6
Workmanship	103	21	42	12	27	90	2	52	10	372	17	2	19	5	3
Logistics	9	3	56	2	21	47	3	30	11	17	207	4	7	2	0
Functionality	7	3	11	6	3	11	1	11	1	2	4	69	3	3	0
Texture	9	4	11	4	11	33	1	38	8	19	7	3	154	2	0
Culture	7	1	3	5	4	11	1	8	3	5	2	3	2	49	0
Festivity	11	7	3	2	8	5	1	8	6	3	0	0	0	0	72

Figure 3. Positive co-occurrence matrix.

From the co-occurrence matrix of positive text (Figure 4), it is evident that consumers' positive emotions primarily stem from perceived dimensions such as appearance design, social aspects, culture, and quality.

Firstly, the co-occurrence frequency of keywords related to the theme of appearance design is notably high, with terms like "exquisite," "like," and "good-looking," indicating that consumers derive aesthetic satisfaction from cultural and creative products. Secondly, the social attributes of these products are also highly recognized, as evidenced by the frequent co-occurrence of terms like "gift," "present," and "friend," highlighting the widespread popularity of cultural and creative products as gifts.

On the other hand, keywords representing the perception of cultural connotation, such as "meaning" and "culture," also show a high frequency. Lastly, the functionality and quality of cultural and creative products are acknowledged, with terms like "quality" and "craftsmanship" reflecting that the product quality meets the expectations of some consumers.

- Negative topic

	Packaging	Appearance	Customer Service	Quality	Defect	Courier	Disappointment	Workmanship	Service	Return	Not worth it	Design	smell	Damage	Discount
Packaging	142	4	3	3	4	17	4	5	5	4	6	3	1	11	4
Appearance	4	44	3	6	5	2	2	6	1	1	3	6	1	3	1
Customer Service	3	3	80	3	2	9	2	2	4	11	2	2	4	2	6
Quality	3	6	3	77	4	3	4	6	1	1	4	5	0	1	2
Defect	4	5	2	4	43	1	1	3	2	0	1	1	0	2	1
Courier	17	2	9	3	1	69	3	2	2	3	0	2	2	5	0
Disappointment	4	2	2	4	1	3	44	3	0	2	4	0	1	1	1
Workmanship	5	6	2	6	3	2	3	48	2	0	5	6	1	1	0
Service	5	1	4	1	2	2	0	2	23	3	5	2	0	0	5
Return	4	1	11	1	0	3	2	0	3	40	1	0	3	4	7
Not worth it	6	3	2	4	1	0	4	5	5	1	61	1	2	0	0
Design	3	6	2	5	1	2	0	6	2	0	1	32	1	0	0
Smell	1	1	4	0	0	2	1	1	0	3	2	1	48	0	0
Damage	11	3	2	1	2	5	1	1	0	4	0	0	0	26	0
Discount	4	1	6	2	1	0	1	0	5	7	0	0	0	0	71

Figure 4. Negative co-occurrence matrix.

The co-occurrence matrix of negative text (see Figure 3) reveals the main sources and reasons for consumers' negative emotions.

Firstly, damaged packaging is identified as one of the primary reasons for consumer dissatisfaction. Secondly, there is a significant disappointment regarding customer service, indicating that consumers expect better support. Additionally, the emergence of negative emotions among some consumers is associated with the craftsmanship of the cultural and creative products. High-frequency co-occurring terms such as "flaws," "craftsmanship," and "not worth it" indicate that certain consumers are disappointed with the quality of the products they purchased.

Moreover, the pricing of cultural and creative products is questioned, as the co-occurrence of terms like "price" and "discount" is notably high, reflecting that some consumers believe these products are not worth their current pricing. Finally, the co-occurrence frequency of "delivery" and "return" indicates that logistical issues directly impact return behaviors..

4. Main findings and design optimization recommendations

4.1. Main findings

The main research findings indicate that cultural and creative products receive widespread positive evaluations in online sales, with positive emotions dominating at 79.4%. Consumers particularly favor appearance design and social gifting attributes. Appearance design emerges as the core focus of consumer interest, with evaluations highly concentrated in this area, emphasizing the importance of aesthetic value and visual appeal in the consumption of cultural and creative products.

The social gifting function and cultural connotations of these products are also significantly valued, suggesting that they are not only material consumer goods but also mediums for cultural transmission and emotional exchange. They serve as social tools, becoming vehicles for conveying emotions and culture in interpersonal relationships.

Negative emotions are primarily concentrated around logistics and customer service, with issues related to packaging, efficiency, and return/exchange timelines directly affecting consumers' purchasing experiences and satisfaction levels. Consumers exhibit a high level of concern regarding the quality and craftsmanship of cultural and creative products, with expectations for high standards in both workmanship and materials.

4.2. Design optimization recommendations

- **Strengthening the Aesthetic Value and Detail of Product Design**

To enhance the aesthetic value of products, it is essential to focus on unique design concepts and exquisite craftsmanship that boost visual appeal. Attention should be given to detail in design, such as incorporating beautiful patterns, color combinations, and high-quality materials. By integrating modern aesthetic trends with traditional elements, products can achieve a balance between cultural depth and contemporary fashion, thereby satisfying consumers' pursuit of aesthetic value.

- **Exploring Cultural Meanings and Deepening Cultural Expression**

The frequent appearance of terms like "meaning" and "imagination" in consumer reviews indicates that when purchasing cultural and creative products, consumers not only value practicality but also seek cultural experiences and emotional satisfaction. Therefore, the design process should emphasize the cultural connotation and expression of the product. By skillfully integrating traditional cultural elements into the design, consumers can appreciate and experience the charm and value of traditional culture through their interaction with the product.

- **Incorporating Creative Elements and Personalized Services in the Design Process**

Consumers have high expectations for the design details and creative elements of cultural and creative products. In the design process, attention should be paid to aspects such as material selection, color coordination, and pattern design to make the products distinctive and high-quality. Additionally, more creative elements can be introduced, such as utilizing AI and AR technologies for engaging interactions. Providing personalized customization services to tailor unique cultural and creative products based on consumer needs can further enhance their appeal.

- **Expression Enhancing Packaging Design and User Experience**

Optimizing packaging design is crucial to ensuring that products are delivered to consumers safely and attractively. Attention should also be paid to logistics efficiency, reasonable pricing, and the quality of customer service to comprehensively enhance the user experience. This holistic approach will improve consumer satisfaction with cultural and creative products, encouraging repeat purchases and positive word-of-mouth.

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